THE INTERNATIONAL UNIVERSITY (IU) – VIETNAM NATIONAL UNIVERSITY – HCMC

Midterm EXAMINATION

Date: .....

Duration: 4500 minutes

Student ID: ..... **Name:.....**

SUBJECT: Experimental Design	
Dean of School of Biotechnology Signature: Full name:	***Lecturer*** Signature:  Full name: Dr Nguyen Tan Khoi

GENERAL INSTRUCTION(S)

1. This is a **open book** examination.
2. **ABSOLUTELY NO** group work!
3. EXCEL spreadsheet is allowed.
4. **No dictionary is allowed.**
5. **No calculator.**

QUESTION

To investigate the effect of the cooling process on the tenderness of meat, 24 pigs were slaughtered and cut in halves (right and left side). One side was then cooled by tunnel cooling (a very quick cooling denoted 'TC' below) and the other side by (conventional) fast cooling (denoted 'FC' below). Furthermore the pigs were divided into two groups: 12 pigs with high pH (ph-group 3) and 12 pigs with low pH (ph-group 2). The parameter associated with tenderness are recorded in the table below.

a. The split-plot design is one of the most common designs we find in experimental design. Explain with sufficient details the reasons why it is common and how this design helps us carrying out the experiment economically and efficiently. (30 pts)

b. Determine if the pork experiment follows the split plot design. If so explain why this design was chosen. Determine the whole-plot and split-plot factors. Explain with sufficient details how you determine these factors. (30 pts)

c. Determine if the cooling process and the pH of the pigs affect the tenderness of meat. (40 pts)

Porker	pH-group	Tunnel	Fast
1	low	7.22	5.56
2	low	3.11	3.33
3	low	7.44	7.00
4	low	4.33	4.89
5	low	6.78	6.56
6	low	5.56	5.67
7	low	7.33	6.33
8	low	4.22	5.67
9	low	3.89	4.00
10	low	5.78	5.56
11	low	6.44	5.67
12	low	8.00	5.33
13	high	8.44	8.44
14	high	7.11	6.00
15	high	6.00	5.78
16	high	7.56	7.67
17	high	5.11	4.56
18	high	8.67	8.00
19	high	5.78	7.67
20	high	6.11	5.67
21	high	7.44	7.56
22	high	7.67	6.11
23	high	8.00	8.22
24	high	8.78	8.44

Good luck!